WEEK 3 · LESSON 14 OF 24 · HANDOUT

Measuring Performance: Accuracy, Precision and Recall

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=Kdsp6soqA7o — open the lesson online to watch inside the course.

Your goal in this lesson: Read a confusion matrix, calculate accuracy, precision and recall, and choose the right measure for a real problem.

Why accuracy is not enough

A mobile-money company has 1,000 transactions; **50 are fraud**. A lazy model that says "not fraud" every time is correct on 950 of them: **95% accuracy** — and it catches **zero** fraud. When one class is rare, accuracy can hide total failure. We need better measures.

The confusion matrix

A better model flags 60 transactions as fraud. Comparing with the truth:

	Really fraud	Really normal
Predicted fraud	True positives (TP) = **40**	False positives (FP) = **20**
Predicted normal	False negatives (FN) = **10**	True negatives (TN) = **930**

- **True positive** — fraud correctly caught.
- **False positive** — a normal transaction wrongly flagged (an annoyed customer).
- **False negative** — fraud that slipped through (lost money).
- **True negative** — normal transaction correctly left alone.

The key measures

Measure	Formula	Our model	Plain question
Accuracy	(TP + TN) ÷ total	(40 + 930) ÷ 1000 = **97%**	How often is it right overall?
Precision	TP ÷ (TP + FP)	40 ÷ 60 = **67%**	When it says "fraud", how often is it really fraud?
Recall	TP ÷ (TP + FN)	40 ÷ 50 = **80%**	Of all real fraud, how much did it catch?
F1 score	2 × P × R ÷ (P + R)	2 × 0.67 × 0.80 ÷ 1.47 ≈ **0.73**	One number balancing precision and recall

Precision or recall — which matters more?

It depends on which mistake is **more costly**:

- **Recall first** when missing a case is dangerous: detecting disease, fraud, or a child at risk of dropping out. Better to check a few extra cases than miss real ones.

- **Precision first** when false alarms are costly: blocking a customer's account, sending a spam warning on an important email, accusing someone.

There is usually a **trade-off**. Flag more cases and recall rises but precision falls; flag fewer and the opposite happens. Many models give a **probability**, and the team chooses a **threshold** (for example, flag if above 0.7) to set the balance.

Measuring regression

For number predictions, we measure how far off the predictions are:

- **MAE (mean absolute error)** — the average size of the error. "Our yield forecast is off by 0.3 tonnes on average."
- **RMSE (root mean squared error)** — like MAE but punishes big errors more.

Always compare with a baseline

A score means little alone. Compare with: the lazy model (always predict the most common answer), a simple rule, or the current human process. "97% accuracy" is weak if the lazy model gets 95%.

Remember this: Use the confusion matrix. Accuracy can mislead on rare events; recall measures how much you catch, precision how trustworthy your alarms are. Choose based on which mistake costs more.

Name: _____ Date: _____

Activity

Task 1

Fill in TP, FP, FN and TN.

Task 2

Calculate accuracy, precision and recall.

Task 3

Which measure matters most here, and why?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Why can accuracy be misleading for fraud detection?
   A. Fraud is always common
   B. Fraud is rare, so always predicting 'not fraud' gives high accuracy while catching nothing
   C. Accuracy cannot be calculated
   D. Accuracy is only for images
2. What is a false negative in fraud detection?
   A. A normal transaction left alone
   B. Fraud that was caught
   C. A normal transaction flagged as fraud
   D. Fraud that the model missed

3. Precision answers which question?

A. Of all real fraud, how much was caught?
B. How fast is the model?
C. How many features are there?
D. When the model says fraud, how often is it right?

4. A model has TP = 40 and FN = 10. What is its recall?

A. 40%
B. 97%
C. 67%
D. 80%

5. For detecting a serious disease, which measure usually matters most?

A. File size
B. Training speed
C. Precision only
D. Recall, because missing a sick patient is dangerous

6. What does MAE measure?

A. The average size of the error in number predictions
B. The model's memory
C. The number of clusters
D. The accuracy of a classifier

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